

# Bronchoscopic pulsed-electric field ablation for carcinoid tumor: robotic-assisted and non-robotic approach

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## ABSTRACT

**Background:** Pulmonary carcinoid tumors represent indolent neuroendocrine neoplasms for which surgery remains the standard of care. Patients ineligible for surgery or radiation have limited treatment options. Endoscopic pulsed-electric field (PEF) ablation via robotic-assisted and non-robotic assisted bronchoscopy platforms is emerging as a minimally invasive alternative for lung malignancies.

**Methods:** A multicenter retrospective study of patients with pulmonary carcinoid lesions treated with bronchoscopic PEF ablation. ssRAB or EBUS were used to navigate to the target lesions and deliver PEF ablation via a 21-gauge needle, tool-in-lesion confirmed by mobile cone beam computed tomography (CT). Procedure details, intra- and post-procedural complications, and follow-up chest CT were assessed for tumor response and safety outcomes.

**Results:** Ten carcinoid tumors from eight patients were successfully ablated. The mean tumor size reduction was 8.10 mm or an overall percentage reduction of 53.9%, range 25–100%. No major complications occurred. Three patients presented mild intraprocedural bleeding. The median follow-up time was 3.5 months (IQR 3–4.5). One patient underwent re-ablation due to tumor recurrence without complications, and two PPN showed complete resolution at follow-up.

**Conclusions:** PEF ablation delivered via ssRAB and EBUS for pulmonary carcinoid tumors appeared feasible and safe, with promising short-term radiographic responses. This approach may provide a minimally invasive treatment option for patients who are not candidates for surgery or radiation.

## 1. Introduction

Lung cancer treatment has undergone substantial advancements over the last 20 years, reflecting major developments in both systemic and localized therapies. As the deadliest cancer for men and women worldwide, continuous efforts are dedicated to optimizing early diagnosis and specialized treatment in order to improve survival outcomes; among them, the introduction of shape-sensing robotic-assisted bronchoscopy (ssRAB) for peripheral pulmonary nodule (PPN) biopsy and endoscopic ablative therapy [1–4].

Shape-sensing robotic-assisted bronchoscopy has maintained

consistent efficacy for PPN sampling throughout its six years of commercial use, supported by reduced complication rates, particularly when compared to image-guided transthoracic biopsy [5–9]. Currently, ssRAB is extending beyond diagnosis, now focusing on delivering endoscopic treatment to localized lung malignancies [3,10]. Pulsed-electric field (PEF) ablation therapy is a non-thermal, minimally invasive technique that employs a monopolar needle to deliver short-duration, high-voltage electrical pulses to a target lesion, inducing programmed cell death while minimizing damage to adjacent tissue [3,11]. Initial applications involved percutaneous access but have recently progressed to ssRAB and endobronchial ultrasound (EBUS)-guided delivery for enhanced

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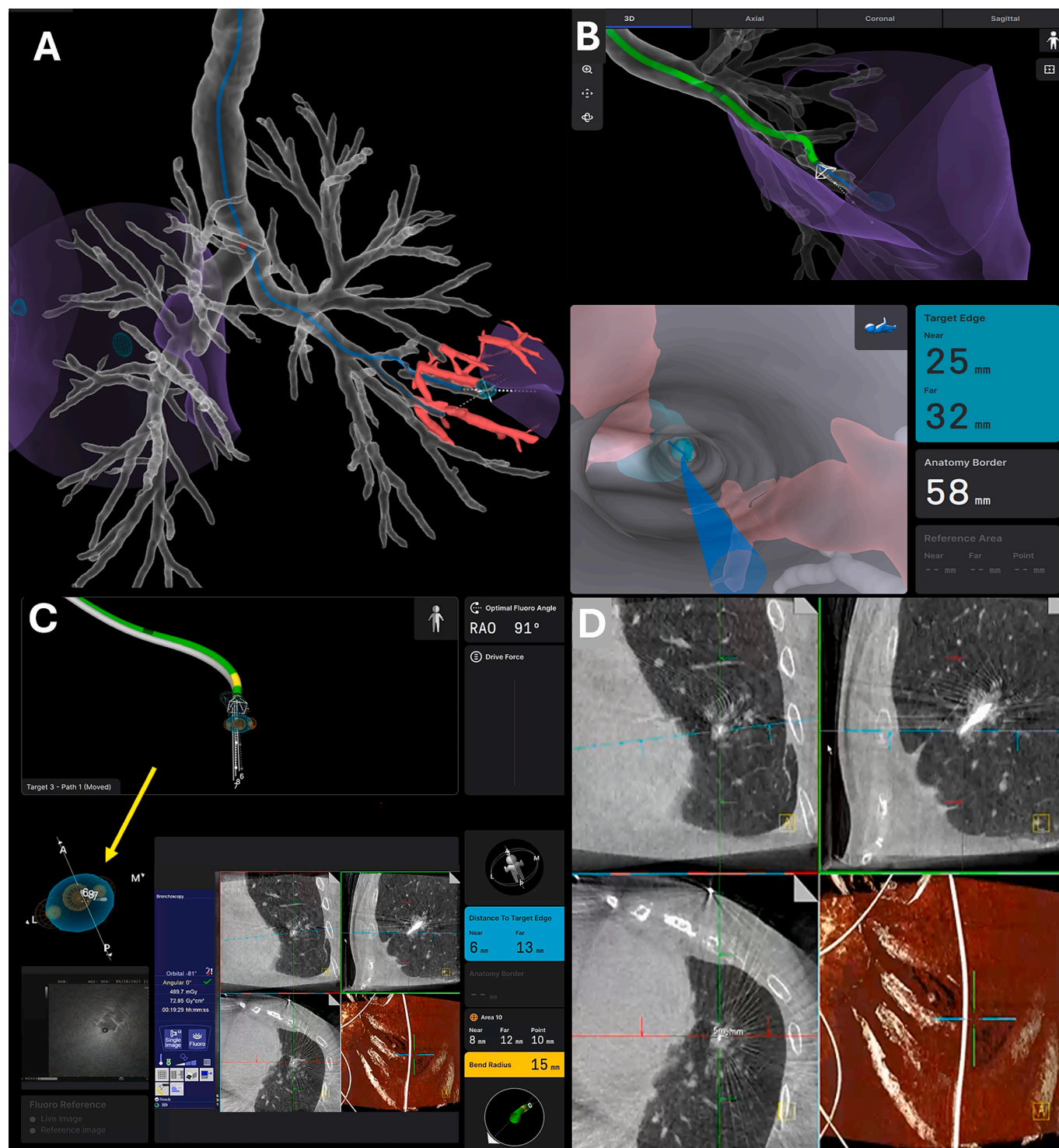
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stability, reach, precision, and safety. Most preliminary data has centered on PEF ablation to non-small cell lung cancer (NSCLC) or metastatic disease, with virtually no reports on its use for other types of lung malignancies such as carcinoid tumors [11–13].

Considered an indolent-behaving neoplasm, typical carcinoid tumors are usually surveilled over time. Management of symptomatic or

progressive disease includes surgery, chemotherapy, and/or somatostatin analogs, while traditional endobronchial techniques, such as laser, electrocautery, argon plasma coagulation, and cryotherapy, provide palliative debulking but risk airway injury or incomplete eradication [14,15]. Henceforth, endoscopic PEF ablation emerges as a potential less-invasive alternative for localized therapy. In this study, we explore



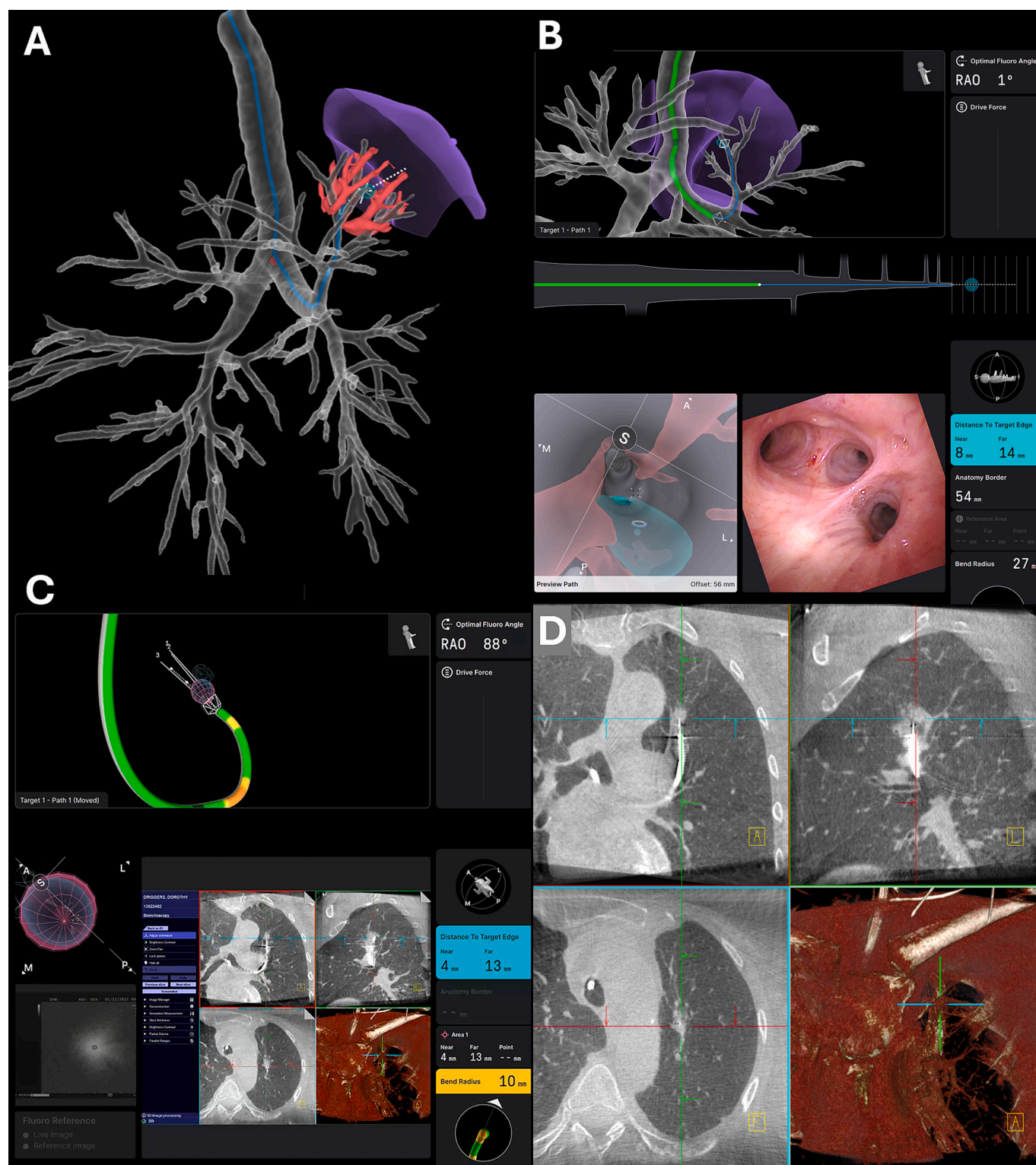
**Fig. 1. Shape-sensing robotic bronchoscopy virtual view.** A-B) Shape-sensing robotic bronchoscopy view of airway reconstruction with target lesion marked in blue, pleura in purple and vascular delimitations in red. C) Vector delimitations in orange on target lesion (yellow arrow), pulsed-electric field ablation (PEF) needle confirmed tool-in-lesion in sagittal, coronal, and axial intraprocedural computed tomography view. D) PEF needle confirmed tool-in-lesion in sagittal, coronal, and axial intraprocedural computed tomography view. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



the feasibility, safety and short-term efficacy of PEF ablation delivered via ssRAB or EBUS for localized pulmonary carcinoid tumor.

## 2. Materials and methods

We conducted a multicenter retrospective cohort study of patients who underwent bronchoscopic PEF ablation delivered by ssRAB or EBUS for treatment of localized pulmonary carcinoid tumor between June



**Fig. 2.** Shape-sensing robotic bronchoscopy virtual view. A-B) Shape-sensing robotic bronchoscopy view of airway reconstruction with target lesion marked in blue, pleura in purple and vascular delimitations in red. C) Marked target nodule margins (yellow arrow), pulsed-electric field ablation (PEF) needle confirmed tool-in-lesion in sagittal, coronal, and axial intraprocedural computed tomography view. D) PEF needle confirmed tool-in-lesion in sagittal, coronal, and axial intraprocedural computed tomography view. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2024 and July 2025. This study was approved and deemed exempt of individual patient consent by the corresponding Institutional Review Board (#25-002566). Adult patients who received PEF ablation for prior or intraprocedural confirmed typical carcinoid tumors, patients with tumoral PPN deemed unsuitable for surgery or radiation, and patients with an established carcinoid diagnosis who declined other invasive treatment options were included in the study. Target lesions under 5 cm and located 10 mm or more from great vessels were considered suitable for endoscopic PEF ablation. Patients who lacked imaging or bronchoscopic follow-up of at least one month post-procedure or in whom PEF ablation was delivered to other non-carcinoid lesions were excluded from the cohort. Demographic and clinical data, PPN characteristics, procedure details, intraprocedural and post-procedural complications, and early radiographic or bronchoscopic outcomes were collected in an encrypted database for statistical analysis.

### 2.1. Bronchoscopic PEF ablation

For ssRAB PEF ablation, anesthesia was induced using rapid sequence intubation with a size 8.5 endotracheal tube under video laryngoscopy, followed by recruitment maneuvers and ventilation with high positive end-expiratory pressure (PEEP) and a tidal volume of 8 mL/kg ideal body weight. Maintenance was achieved with total intravenous anesthesia (TIVA) using propofol and neuromuscular blockade with rocuronium. Standard American Society of Anesthesiologists (ASA) monitoring, quantitative electroencephalographic, and neuromuscular monitoring, were employed to ensure adequate anesthetic depth and optimal procedural conditions. Under general anesthesia, standard flexible bronchoscopy was used to inspect the airway for any abnormalities or endobronchial lesions and clear secretions. ssRAB was then performed using the Ion Endoluminal System™ (Intuitive Surgical Inc., Sunnyvale, CA, USA) with computer-generated image-guided navigation and radial EBUS and mobile cone-beam CT (mCBCT) with 3-dimensional (3D) fluoroscopy to confirm tool-in-lesion, including vasculature delimitations (Figs. 1 and 2). In cases where the target lesion and/or other lesions required sampling for diagnostic confirmation, biopsy was performed first, using fine needle aspiration, forceps, or the 1.1 mm cryoprobe at the discretion of the proceduralist. Rapid on-site pathology evaluation (ROSE) was available for assessing sample adequacy and diagnosis. Once the target carcinoid tumors were confirmed, the Aliya® PEF system (Galvanize Therapeutics Inc, Redwood City, CA, USA) was used to deliver PEF ablation. A 21G INUMI Flex Needle™ (Galvanize Therapeutics Inc, Redwood City, CA, USA) was introduced towards PPN with tool-in-lesion confirmed by mCBCT. The number of vectors and activations, each lasting a standard five minutes, was dependent on the lesion characteristics and proceduralist preference. Intraprocedural complications such as bleeding were managed in a case-by-case scenario. Endobronchial ultrasound transbronchial needle aspiration (EBUS-TBNA) was completed for hilar and mediastinal staging when deemed necessary. Patients were admitted overnight for observation, chest X-ray evaluation, and discharged according to institutional protocols.

EBUS-guided PEF ablation followed similar anesthesia and procedural stepwise process, using the Olympus curvilinear endobronchial ultrasound bronchoscope (Olympus BF-UC190F, Olympus Corporation, Tokyo, Japan) for biopsy and the 21G INUMI Flex Needle™ for PEF ablation delivery.

### 2.2. Variables of interest

Relevant demographic and clinical information were recorded, including prior history of malignancy and pulmonary comorbidities. For carcinoid PPNs ablated, we collected lobar location, size, and type. Size was documented in two and three dimensions, depending on institutional practices. Nodule type was categorized based on density on CT imaging as solid, subsolid (ground-glass and part-solid), cystic/cavitary

or mixed (combination of densities other than part-solid). If more than one PPN was ablated in the same procedure or if the same PPN was re-ablated, they were considered as separate, individual ablated lesions.

Procedural details comprised concurrent biopsy and PEF ablation or PEF ablation alone, duration of procedure, total length of stay in days, and complications. Adverse events were recorded intraprocedurally and within seven days of the ablation, including bleeding, pneumothorax, or any other event requiring intervention or escalation of care. Major adverse events were considered as grade  $\geq 3$  bleeding according to the Nashville Bleeding Scale [16] for bronchoscopic procedures, pneumothorax requiring chest tube, and grade  $\geq 3$  in the Common Terminology Criteria for Adverse Events (CTCAE) [17] up to seven days post-procedure.

### 2.3. Statistical analysis

Descriptive statistics were used to summarize the data. Continuous variables were presented as median and interquartile range (IQR), categorical variables were expressed as counts and percentages. Statistical analysis was performed using SPSS 26.0 (IBM, 2019).

## 3. Results

Ten pulmonary carcinoid tumors from eight patients were ablated with PEF delivered via ssRAB or EBUS (Table 1). The median age was 76 (IQR 73.5–78) years, the mean BMI was 28.9 kg/m<sup>2</sup> ( $\pm$ SD 5.6), all patients were female, six were former smokers, and four had a prior history of malignancy.

Eight PPN from seven patients were ablated with ssRAB, and two PPN from one patient were ablated using linear EBUS. Five patients underwent bronchoscopic PEF alone, after prior diagnostic confirmation. Three patients underwent PPN biopsy and PEF ablation, while two patients underwent PPN biopsy, PEF ablation, and EBUS-TBNA for hilar and mediastinal staging during the same procedure (Table 1).

The median maximum diameter of PPN pre-PEF ablation was 13.3 mm (IQR 8.0–21.0), 40% ( $n = 4$ ) were located in the LLL and all were solid in density on CT. One carcinoid tumor presented endobronchially; the remaining nine lesions were located peripherally. The median procedure duration was 129 mins (IQR 85–177) with a median of 4 activations (IQR 3.0–4.5) of 5 min each (Table 1). Two instances of intraprocedural bleeding were reported, one grade 2 managed with bronchoscope wedging and one grade 3 controlled with Fogarty balloon compression. No pneumothorax or other complications were reported within seven days of the intervention. The total length of stay was one day as all patients were discharged after overnight admission for observation.

The median time to follow-up was 3.5 months (IQR 3.0–4.5). The median maximum diameter of PPN post-PEF ablation was 6.9 mm (IQR 3.3–9.3), exhibiting a mean absolute reduction in size of 8.10 mm or, proportionally, a reduction in size of 59.3%, ranging from 25% to 100% (Figs. 3 and 4). Two patients showed complete resolution, one radiographically and the other bronchoscopically (Fig. 3. D and Fig. 4. B) and in one patient, two non-treated lesions showed apparent “abscopal” effect with interval volume reduction of 24.3% on CT.

## 4. Discussion

Pulsed-electric field ablation delivered through ssRAB and EBUS for localized treatment of ten carcinoid tumors in eight patients was feasible and safe in this exploratory study. Endoscopic PEF ablation resulted in a mean absolute reduction in size of 8.10 mm or, proportionally, 53.9% overall, ranging from 25% to 100%, with no major immediate or post-procedural complications (Table 1).

As a minimally invasive non-thermal ablative technique, PEF ablation was cleared by the FDA in 2024 to treat solid tumors using short-high voltage pulses [13,18,19]. Currently, data regarding endoscopic



**Table 1**

Bronchoscopic pulsed-electric field ablation procedure details and follow-up.

| Case No. | Target Location | Nodule Density | Bronchoscopy Technique | Procedure Performed                       | Pre-PEF Ablation Size (mm) | Duration (mins) | No. Activations/Duration (mins) | Post-PEF Ablation Size (mm) | Follow Up (months) |
|----------|-----------------|----------------|------------------------|-------------------------------------------|----------------------------|-----------------|---------------------------------|-----------------------------|--------------------|
| 1        | RUL             | Solid          | ssRAB                  | 3 PPN biopsy + 1 PEF ablation + EBUS-TBNA | 8.5 × 6.8                  | 180             | 4 / 5                           | 3.9 × 2.7                   | 12                 |
| 2        | LLL             | Solid          | ssRAB                  | 3 PPN biopsy + 2 PEF ablation*            | 8.0                        | 172             | 4 / 5                           | 0.0**                       | 6                  |
| 3        | LLL             | Solid          | ssRAB                  | 1 PPN biopsy + 1 PEF ablation + EBUS-TBNA | 8.5 × 8.0                  | 177             | 1 / 5                           | 6.6 × 5.8                   | 2                  |
| 4        | LUL             | Solid          | ssRAB                  | PEF ablation                              | 7.7 × 7.0                  | 86              | 2 / 5                           | 5.4 × 5.0                   | 3                  |
| 5        | RLL             | Solid          | ssRAB                  | PEF ablation                              | 14.5 × 13.0                | 85              | 4 / 5                           | 11.5 × 7.0                  | 4                  |
| 6        | RUL             | Solid          | ssRAB                  | PEF ablation                              | 30.0                       | 26              | 3 / 5                           | 0.0**                       | 1                  |
| 7        | LLL             | Solid          | Linear EBUS            | PEF ablation                              | 34.0 × 29.0 × 20.0         | 93              | 12 / 5                          | 13.0 × 18.0 × 12.0          | 3                  |
|          | LLL             | Solid          | Linear EBUS            | PEF ablation                              | 26.0 × 13.0 × 24.0         | 79              | 8 / 5                           | 21.0 × 10.0 × 16.0          | 3                  |
| 8        | RLL             | Solid          | ssRAB                  | PEF ablation                              | 16.0 × 12.0 × 13.0         | 84              | 4 / 5                           | 10.0 × 7.0 × 6.0            | 4                  |
|          | RLL             | Solid          | ssRAB                  | PEF ablation                              | 13.0 × 13.0 × 13.0         |                 | 4 / 5                           | 9.0 × 7.0 × 7.0             | 4                  |

ssRAB = shape-sensing robotic-assisted bronchoscopy, EBUS = endobronchial ultrasound, TBNA = transbronchial needle aspiration, PPN = peripheral pulmonary nodule, PEF = pulsed-electric field, RUL = right upper lobe, LLL = left lower lobe, LUL = left upper lobe, RLL = right lower lobe.

\* For case 2, the patient underwent a second concurrent ablation for another non-carcinoid tumor lung malignancy.

\*\* For case 2, follow-up computed tomography showed scarring, with no evidence of the LLL lesion; for case 6, follow-up bronchoscopy showed no evidence of endobronchial tumor.

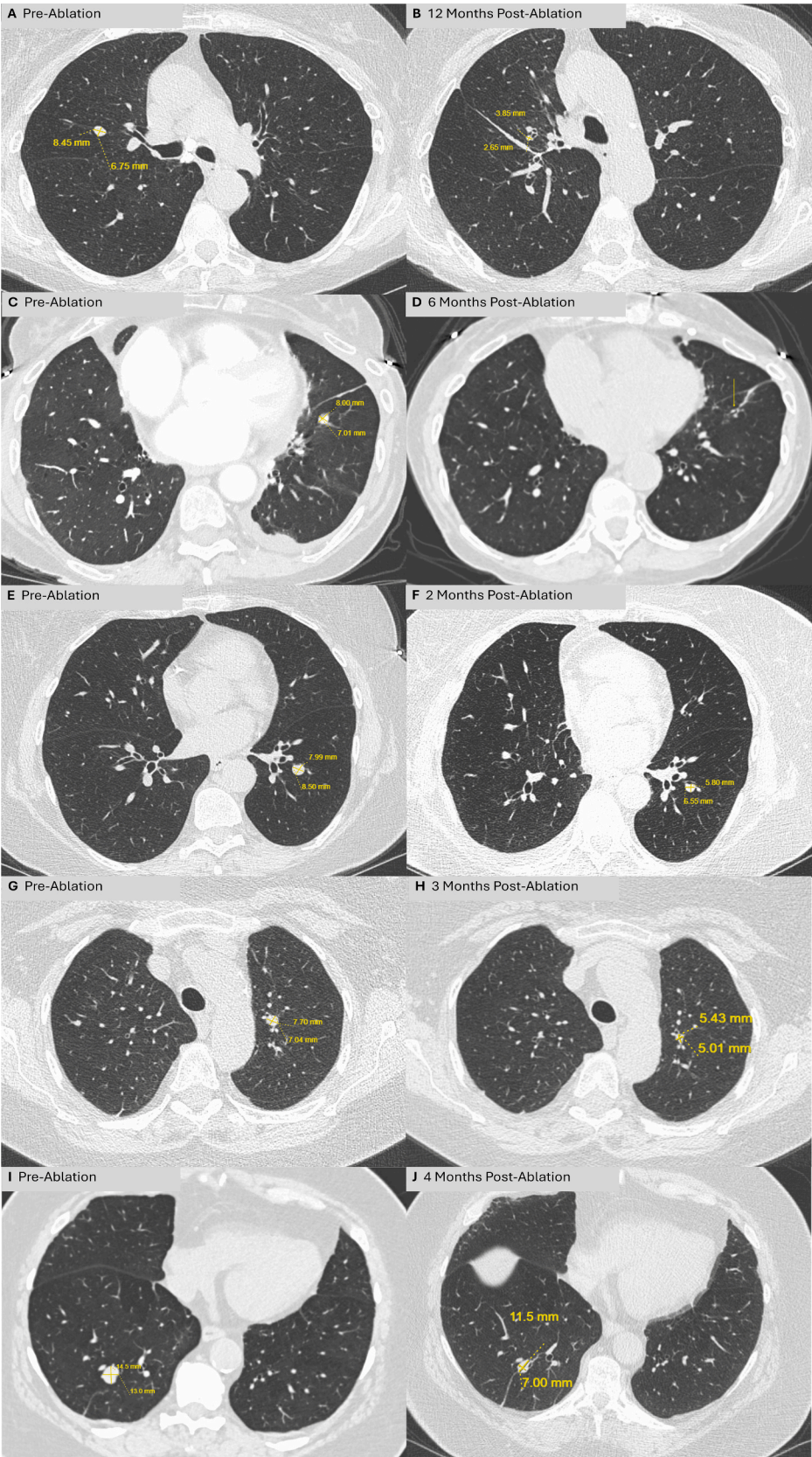
PEF ablation comprise preliminary findings from animal studies, early commercial use, and clinical trials [19]. Safety and feasibility were assessed in the INCITE ES trial (NCT04732520) for pre-surgical PEF ablation for stage I-IB NSCLC, where 26 out of 36 patients received endoscopic ablation, while the remaining 10 underwent percutaneous ablation. Minor adverse events were reported as pneumothoraces, hemoptysis, and sore throat. Notably, post-surgical tissue analysis revealed PEF-induced cellular tumor depleted or absent zones [11]. In our study, patient #6 underwent bronchoscopic biopsy of the ablated RUL lesion one month post-intervention, which revealed tumor depleted zones comprising benign respiratory mucosa and granulation tissue. Recently, the AFFINITY trial (NCT05890872) assessed endoscopic delivery of PEF ablation to target stage IV NSCLC or lung metastatic disease for ablative neoadjuvant therapy prior to first-line treatments. Out of the 30 patients enrolled, 27 received PEF ablation through ssRAB and 3 through percutaneous approach [20]. Their outcomes describe the first reports on PEF ablation delivered through ssRAB. The procedural success rate was 100% and did not alter subsequent first-line treatment schedule. Moreover, the only procedure-related adverse effect, a pneumothorax requiring chest tube, occurred in the percutaneous group, reinforcing the safety advantage of ssRAB over transthoracic techniques. Similarly, no serious adverse events occurred in our cohort, even when two lesions were ablated within the same procedure such as case #8. Active bleeding was controlled intraprocedurally using Fogarty balloon compression in one patient and bronchoscope wedging in a second one, without subsequent complications. In two additional patients, Fogarty balloon compression was applied prophylactically and no bleeding was observed throughout the procedure, as described in a prior case series on preventive measures for bleeding control during robotically-assisted bronchoscopic ablation [21].

Procedure times were relatively long, particularly in the first three cases, due to several procedural and technical factors. Each ablation required 5-minute activations, with a median of four activations per lesion, and cone-beam CT confirmation of vector and needle positioning before each activation, adding to the overall duration. The number of activations and needle repositioning varied according to the number, size, and location of target lesions. In addition, biopsies were performed prior to ablation in three of eight cases, and in two, EBUS-TBNA was conducted for mediastinal and hilar staging during the same session,

further prolonging procedure time. However, when ablation was performed alone, duration was under 90 min in most cases and under 30 min in one case, more closely consistent with reported times for other bronchoscopic interventions. Procedural duration was influenced by the early stages of the learning curve and with increased operator experience and workflow optimization, such as refining needle positioning, vector markings, and activation settings, potential intraprocedural delays are expected to be reduced.

Conventional ablative techniques include thermal and non-thermal approaches. Microwave, radiofrequency, laser, and cryoablation, achieve tumor destruction through temperature-mediated mechanisms and have demonstrated efficacy in select pulmonary malignancies; however, their application may be limited by collateral thermal injury, heat-sink effects, and inconsistent ablation margins, particularly for centrally located or peribronchial lesions adjacent to critical vascular and airway structures [19]. Percutaneous delivery of thermal ablative techniques is additionally associated with a substantial risk of pneumothorax and pleural injury [19,22,23]. Cryoablation delivered by ssRAB has been recently implemented to treat NSCLC in three patients, without procedure-related complications [24]. However, it is hindered by long freeze-thaw cycles of up to 25 min per ablation [19,25]. Non-thermal techniques like photodynamic therapy (PDT) and PEF achieve tumor destruction without relying on temperature-mediated cytotoxicity, thereby reducing collateral injury to surrounding bronchovascular structures. PDT is used extensively in central airway tumors, although its limitations include shallow tissue penetration, dependence on precise fiber positioning, and photosensitivity-related adverse effects [26,27]. In contrast, PEF induces non-thermal cell death through irreversible electroporation and appears to enhance local and distant antitumor immune responses, without evidence to date of major adverse events beyond those typically associated with the delivery approach [12,13]. As such, PEF ablation may serve as a minimally invasive option for endobronchial, peripheral, and central tumors for which other targeted treatment options may be suboptimal.

To our knowledge, this is the first publication on robotic-assisted and non-robotic bronchoscopy PEF ablation for carcinoid tumors. Part of a larger family of rare malignant neuroendocrine tumors, pulmonary carcinoids comprise typical (85–90%) and atypical subtypes (10–15%) [14]. The indolent, low-grade nature of typical carcinoids is suitable for



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**Fig. 3. Pre- and post-pulsed-electric field (PEF) ablation.** A) Case 1: chest computed tomography (CT) axial view of  $8.45 \times 6.75$  mm right upper lobe (RUL) solid lesion prior to endoscopic PEF ablation. B) Case 1: chest CT axial view of  $3.85 \times 2.65$  mm RUL solid lesion 12 months after endoscopic PEF ablation. C) Case 2: chest CT axial view of  $8.00 \times 7.01$  mm left lower lobe (LLL) solid lesion prior to endoscopic PEF ablation. D) Case 2: chest CT axial view showing scarring tissue in the LLL treated lesion 6 months after endoscopic PEF ablation. E) Case 3: chest CT axial view of  $8.50 \times 7.99$  mm LLL solid lesion prior to endoscopic PEF ablation. F) Case 3: chest CT axial view of the  $6.55 \times 5.80$  mm LLL solid lesion 2 months after endoscopic PEF ablation. G) Case 4: chest CT axial view of  $7.04 \times 7.00$  mm left upper lobe (LUL) solid lesion prior to endoscopic PEF ablation. H) Case 4: chest CT axial view of  $5.43 \times 5.01$  mm LUL solid lesion 3 months after endoscopic PEF ablation. I) Case 5: chest CT axial view of  $14.5 \times 13.00$  mm RLL solid lesion prior to endoscopic PEF ablation. J) Case 5: chest CT axial view of  $11.5 \times 7.00$  mm RLL solid lesion 4 months after endoscopic PEF ablation.

surveillance in the absence of symptoms or metastasis. Recent developments in therapy for advanced stages include somatostatin analogs, antiproliferative agents, chemotherapy, and immune checkpoint therapy [15,28]. For localized disease, surgery is the treatment of choice, as carcinoid tumors are less responsive to radiotherapy than other neuroendocrine malignancies [14]. In our study, patients who underwent bronchoscopic PEF ablation were either not candidates for surgery or radiation due to high-risk comorbidities or had declined invasive surgical interventions. In addition, lesions under 5 cm and located less than 10 mm from great vessels were considered suitable for ablation, following prior reports demonstrating feasibility and safety in other malignant thoracic lesions with these characteristics [20]. All treated patients demonstrated a favorable early response, evidenced by radiographic reduction in tumor size and/or resolution, and, in one case, complete absence of visible endobronchial tumor on bronchoscopy, within a follow-up period of 1–4 months. Two carcinoid lesions showed sustained positive response at 6 to 12-month follow-up, and, in patient #8, an apparent abscopal effect was observed when two non-treated nodules also decreased in size in subsequent surveillance imaging. Only one patient required retreatment within the first three months.

Furthermore, two patients underwent ssRAB diagnosis, PEF ablation, and EBUS-TBNA for staging within the same anesthetic event, followed by surgical resection and/or radiotherapy on subsequent dates for concurrent lung primary adenocarcinomas. Potential discrepancies between intraprocedural diagnosis and final pathology should be considered, and caution is warranted when on-site biopsy evaluation is inconclusive. In our study, the diagnosis of carcinoid tumor in the two target PPN biopsied prior to ablation was determined intraprocedurally and confirmed on final pathology. Other ablative techniques have been described concurrently with surgical resection, such as a case report on laser ablation and robotic-assisted segmentectomy to treat contralateral adenocarcinomas, while laser and cryoablation were used in sequence with high-frequency electrocoagulation to manage a bronchial schwannoma causing obstruction [29,30]. In both cases, no intra- or post-procedural adverse events were noted. This multimodal approach encompassing diagnostic bronchoscopy, different endoscopic ablative techniques, and EBUS-TBNA staging allowed specialized care for multiple lung primary and metastatic malignancies, tailored to specific presentation, patients' functional status and preferences. In this manner, ssRAB and interventional pulmonology techniques are becoming the "one-stop-shop" for diagnosis, staging, and treatment for suspicious PPN.

We acknowledge the limitations of our study. First, its retrospective nature is subject to selection bias. Second, as a feasibility and safety study, we included a small cohort of eight patients and ten target PPN, which may not accurately represent the overall population of carcinoid tumor patients. Lastly, our results largely entail short-term follow-up ranging from 1 to 4 months post-procedure, with only two cases followed for 6 to 12 months, for an indolent malignancy that is usually followed over extended periods. The lack of long-term data and the small cohort limit the validation and extrapolation of our results. Continued research is necessary to define parameters for optimal use, including number and duration of pulses, probe accessibility and location confirmation, as well as anesthesia requirements. More importantly, long-term clinical and radiographic follow-up, need and timing of retreatment, larger multicenter cohorts, and comparative trials with other localized treatment approaches are warranted to comprehensively assess the efficacy of PEF ablation as a minimally invasive therapy for

pulmonary carcinoid tumors [10].

Shape-sensing robotic-assisted bronchoscopy and EBUS-guided PEF ablation was feasible and safe for the localized treatment of ten peripheral carcinoid tumors in eight patients. These short-term preliminary findings are encouraging for the advancement of endoscopic localized treatment, contributing to bronchoscopy techniques as single diagnostic, staging, and treatment option for pulmonary lesions suspicious thoracic malignancies.

#### Disclosure statement

Two patients, patients #3 and #4, are included in a previously published case series focused on preventive measures for bleeding control during robotically-assisted bronchoscopic ablation: Fernandez-Bussy S, Kornfeld A, Husta B, Vu LH, Lee-Mateus AY, Alqawasma MA, Abia-Trujillo D. AIM (Anticipate/Adjust, Image, Manage) to Prevent Bleeding During Robotically-Assisted Bronchoscopic Ablation. *Respirol Case Rep.* 2025 Sep 4;13(9):e70335. <https://doi.org/10.1002/rcr2.70335>. PMID: 40917250; PMCID: PMC12411101.

#### CRediT authorship contribution statement

**Sebastian Fernandez-Bussy:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Alejandra Yu Lee-Mateus:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Regina M. Murillo-Torres:** Writing – review & editing, Data curation. **Paola Gutierrez-Gallegos:** Writing – review & editing, Data curation. **Bryan Husta:** Writing – review & editing, Methodology. **Linh H. Vu:** Writing – review & editing, Methodology. **Ryan M. Kern:** Writing – review & editing, Methodology. **John Mullon:** Writing – review & editing, Methodology. **Justin M. Thomas:** Writing – review & editing, Methodology. **Zachary J. Fleissner:** Writing – review & editing, Methodology. **Kelly S. Robertson:** Writing – review & editing, Resources. **Britney N. Hazelett:** Writing – review & editing, Resources. **Ellen Love:** Writing – review & editing, Project administration. **David Abia-Trujillo:** Writing – review & editing, Methodology, Conceptualization.

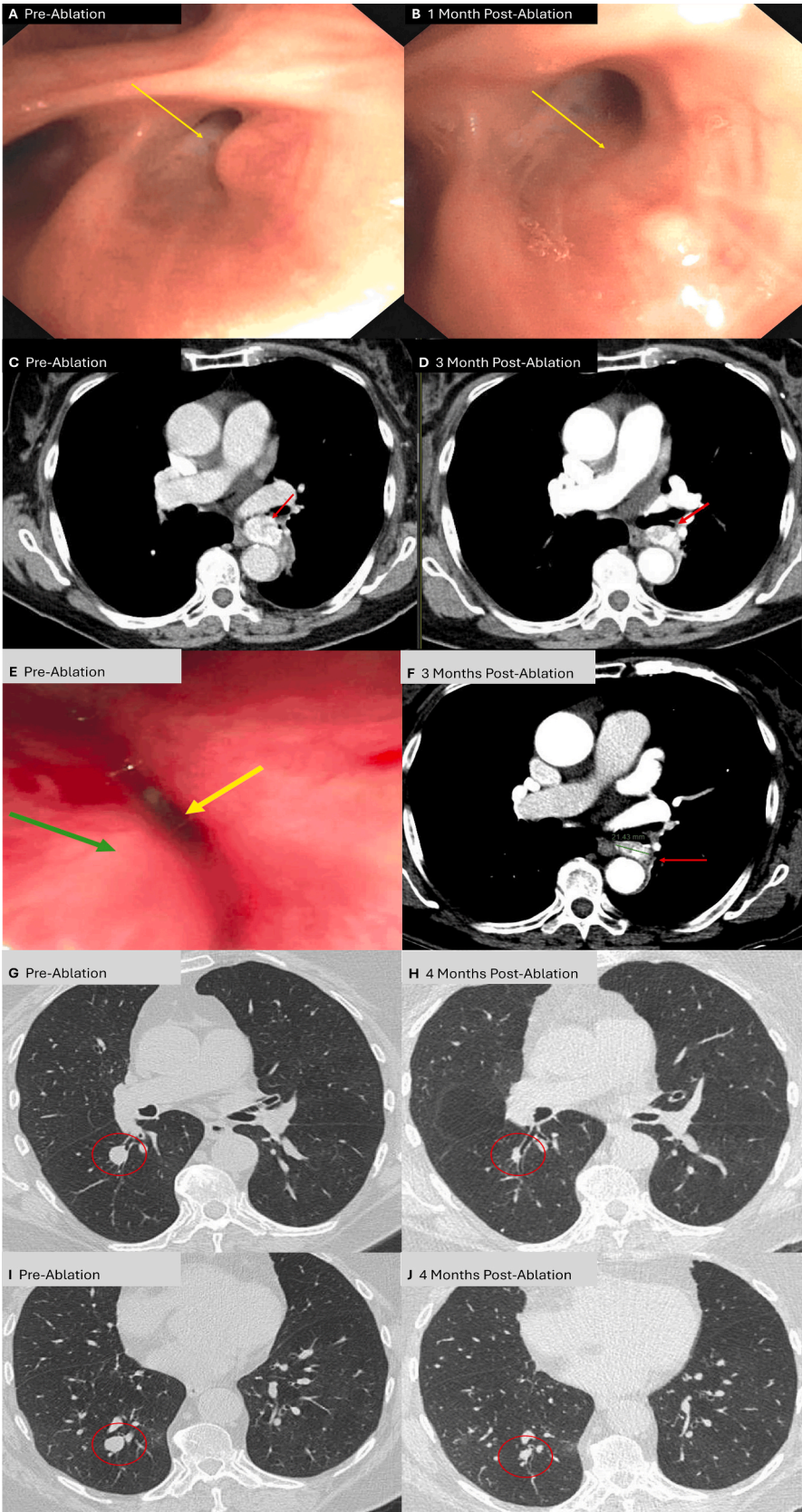
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#### Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sebastian Fernandez-Bussy, MD reports a relationship with Intuitive Surgical Inc that includes: consulting or advisory, speaking and lecture fees, and travel reimbursement. David Abia-Trujillo, MD reports a relationship with Intuitive Surgical Inc that includes: consulting or advisory, speaking and lecture fees, and travel reimbursement. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.





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**Fig. 4. Pre- and post-pulsed-electric (PEF) field ablation.** A) Case 6: bronchoscopy view of 30 mm endobronchial tumor in right upper lobe (RUL) prior to endoscopic pulsed-electric field (PEF) ablation (yellow arrow). B) Case 6: bronchoscopy view of RUL 1 month after endoscopic PEF-ablation, showing absence of tumoral tissue (yellow arrow). C) Case 7: chest CT axial view of 34.00 × 29.00 × 20.00 mm left hilar mass completely obstructing the left lower lobe (LLL) bronchus prior to endoscopic PEF ablation (red arrow). D) Case 7: chest CT axial view of 13.00 × 18.00 × 12.00 mm reduced left hilar mass 3 months after to endoscopic PEF ablation (red arrow). E) Case 7: bronchoscopy view of 26.00 × 13.00 × 24.00 mm of regressed left hilar mass prior to second PEF ablation (green arrow). F) Case 7: chest CT axial view of 21.0 × 10.0 × 16.0 mm of reduced left hilar mass 3 months after PEF re-ablation (red arrow). G) Case 8: right lower lobe (RLL)#1 chest CT axial view of 16.00 × 12.00 × 13.00 mm RLL#1 solid lesion prior to endoscopic PEF ablation (red circle). H) Case 8: RLL#1 chest CT axial view of 10.00 × 7.00 × 6.00 mm RLL#1 solid lesion 4 months after endoscopic PEF ablation (red circle). Case 8: I) Case 8: RLL#2 chest CT axial view of 13.00 × 13.00 × 13.00 mm RLL#2 solid lesion prior to endoscopic PEF ablation (red circle). J) Case 8: RLL #2 chest CT axial view of 9.00 × 7.00 × 7.00 mm RLL#2 solid lesion 4 months after endoscopic PEF ablation (red circle). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

## Data availability

Data are available upon reasonable request. The data that support the findings of this study are available from the corresponding author (SFB), upon reasonable request.

## References

- [1] R.L. Siegel, T.B. Kratzer, A.N. Giaquinto, H. Sung, A. Jemal, Cancer statistics, 2025, *CA Cancer J. Clin.* 75 (1) (2025) 10–45.
- [2] P.-L. Su, N. Furuya, A. Asrar, et al., Recent advances in therapeutic strategies for non-small cell lung cancer, *J. Hematol. Oncol.* 18 (1) (2025) 35.
- [3] J.W.Y. Chan, I.C.H. Siu, A.T.C. Chang, et al., Review on endobronchial therapies: current status and future, *Ann Transl Med* 12 (4) (2024) 75.
- [4] S. Fernandez-Bussy, N.C. Chandra, A. Koratala, et al., Robotic-assisted bronchoscopy: a narrative review of systems, *J. Thorac. Dis.* 16 (8) (2024) 5422–5434.
- [5] P. Balasubramanian, D. Abia-Trujillo, A. Barrios-Ruiz, et al., Diagnostic yield and safety of diagnostic techniques for pulmonary lesions: systematic review, meta-analysis and network meta-analysis, *Eur. Respir. Rev.* 33 (173) (2024) 240046.
- [6] A.R. Brownlee, C. Perez, L. Weiser, et al., 1121 Shape-sensing Robotic-assisted Bronchoscopic Biopsies: Diagnostic Yield and Surgical Implications. The Annals of Thoracic Surgery [Internet] 2025 [cited 2025 July 8]; Available from: <https://www.sciencedirect.com/science/article/pii/S000349752500325X>.
- [7] S. Fernandez-Bussy, A. Yu Lee-Mateus, A. Barrios-Ruiz, et al., Diagnostic performance of shape-sensing robotic-assisted bronchoscopy for pleural-based and fissure-based pulmonary lesions, *Thorax* 80 (3) (2025) 150–158.
- [8] A. Yu Lee-Mateus, J. Reisenauer, J.C. Garcia-Saucedo, et al., Robotic-assisted bronchoscopy versus CT-guided transthoracic biopsy for diagnosis of pulmonary nodules, *Respirology* 28 (1) (2023) 66–73.
- [9] S. Fernandez-Bussy, A. Yu Lee-Mateus, J. Reisenauer, et al., Shape-sensing robotic-assisted bronchoscopy versus computed tomography-guided transthoracic biopsy for the evaluation of subsolid pulmonary nodules, *Respiration* 103 (5) (2024) 280–288.
- [10] T.C. Chopko, J.W. Maroun, P.W. Eiken, K.N. Lee, C.J. Reisenauer, J.S. Reisenauer, Fire or ice - emerging therapies for unresectable pulmonary nodules, *Int. J. Surg.* 111 (8) (2025) 5350–5362.
- [11] M. Jimenez, J. Flandes, E.H.F.M. van der Heijden, et al., Safety and feasibility of pulsed electric field ablation for early-stage non-small cell lung cancer prior to surgical resection, *J. Surg. Oncol.* 131 (8) (2025) 1529–1542.
- [12] M. Jimenez, J.M. Fernandez, W.S. Krinsky, Pulsed Electric Field (PEF) treatment of progressive non-small cell lung cancer concurrently treated with immune checkpoint blockade: a case report, *Respir. Med. Case. Rep.* 49 (2024) 102018.
- [13] B.M. Arciga, D.M. Walters, E.T. Kimchi, et al., Pulsed electric field ablation as a candidate to enhance the anti-tumor immune response to immune checkpoint inhibitors, *Cancer Lett.* 609 (2025) 217361.
- [14] F. Limaem, M.A. Tariq, U. Ismail, J.M. Wallen, Lung Carcinoid Tumors [Internet], in: StatPearls. Treasure Island (FL): StatPearls Publishing; 2025 [cited 2025 July 21]. Available from: <http://www.ncbi.nlm.nih.gov/books/NBK537080/>.
- [15] N. Subash, M.M. Papali, K.P. Bahadur, et al., Recent advances in the diagnosis and management of carcinoid syndrome, *Dis. Mon.* 68 (7) (2022) 101304.
- [16] E.E. Folch, A.K. Mahajan, C.L. Oberg, et al., Standardized definitions of bleeding after transbronchial lung biopsy: a Delphi consensus statement from the nashville working group, *Chest* 158 (1) (2020) 393–400.
- [17] CTCAE and AE Reporting - NCI [Internet]. 2025 [cited 2026 Feb 5]; Available from: <https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events>.
- [18] M.R. Rouhezamin, F.J. Fintelmann, A.J. Huang, et al., Limited effectiveness in early human clinical experience with pulsed electrical field ablation, *J. Vasc. Interv. Radiol.* 36 (2) (2025) 274–281.
- [19] S. Fernandez-Bussy, P. Gutierrez-Gallegos, A. Yu Lee-Mateus, et al., Endoscopic ablation for malignant lung lesions: current techniques, unmet needs, and future directions, *Respiration* (2025) 1–16.
- [20] M.A. Pritchett, J.S. Reisenauer, S. Fernandez-Bussy, et al., The safety of pulsed electric field ablation before standard of care treatment for patients with metastatic cancer, *J. Bronchology Interv Pulmonol* 32 (4) (2025) e01027.
- [21] S. Fernandez-Bussy, A. Kornfeld, B. Husta, et al., AIM (Anticipate/Adjust, Image, Manage) to prevent bleeding during robotically-assisted bronchoscopic ablation, *Respirol Case Rep* 13 (9) (2025) e70335.
- [22] B.T. Welch, W. Brinjikji, G.D. Schmit, et al., A national analysis of the complications, cost, and mortality of percutaneous lung ablation, *J. Vasc. Interv. Radiol.* 26 (6) (2015) 787–791.
- [23] M. Kashima, K. Yamakado, H. Takaki, et al., Complications after 1000 lung radiofrequency ablation sessions in 420 patients: a single center's experiences, *AJR Am. J. Roentgenol.* 197 (4) (2011) W576–W580.
- [24] C. Zhang, F. Xie, H. Xi, et al., Shape-sensing robotic-assisted bronchoscopy combined with cone-beam computed tomography-guided cryoablation for malignant lung tumors, *Respiration* 104 (12) (2025) 963–973.
- [25] J.L. Hinshaw, P.J. Littrup, N. Durick, et al., Optimizing the protocol for pulmonary cryoablation: a comparison of a dual- and triple-freeze protocol, *Cardiovasc. Intervent. Radiol.* 33 (6) (2010) 1180–1185.
- [26] H. Chang, Y.-C. Chiu, S.-W. Lee, et al., Novel light delivery method for performing transbronchial photodynamic therapy ablation to treat peripheral lung cancer: a pilot study, *Photodiagn. Photodyn. Ther.* 40 (2022) 103063.
- [27] H. Chang, K.S. Liao, Y.-S. Hsieh, Bronchoscopic light delivery method for peripheral lung cancer photodynamic therapy, *J. Thorac. Dis.* 12 (7) (2020) 3611–3621.
- [28] M. Rutherford, M. Wheless, K. Thomas, R.A. Ramirez, Current and emerging strategies for the management of advanced/metastatic lung neuroendocrine tumors, *Curr. Probl. Cancer* 49 (2024) 101061.
- [29] T. Sun, Y. Ge, Z. Chen, T. Ma, S. Feng, H. Zhang, Ion robotic bronchoscopy laser ablation and Da Vinci robotic segmentectomy for bilateral pulmonary nodules: a case report, *Transl Lung Cancer Res* 14 (2) (2025) 619–624.
- [30] X. Li, X. Yu, R. Luo, X. Wang, Interventional treatment of right lower lobe bronchial schwannoma under bronchoscopy: a case report and literature review, *Sci. Prog.* 107 (4) (2024) 00368504241305911.